Introduction to the Architecture of Large Language Models

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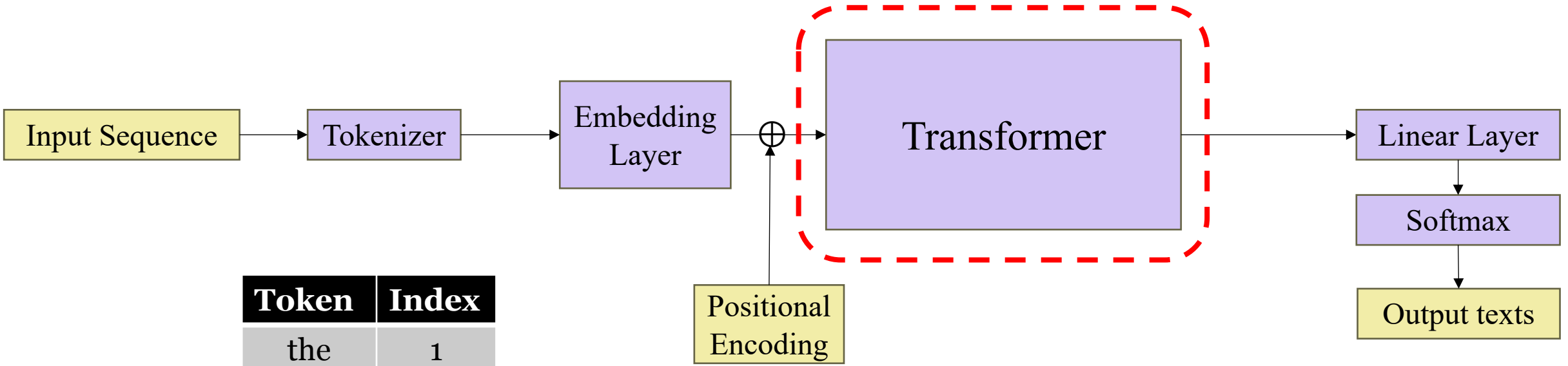
Content

- Overview of the LLM Architecture
- Basic Components: Self-Attention and FFN
- Serving Process of Decoder-only LLM: Prefill vs. Decode
- Infrastructure for LLM Serving
- Common Challenges in LLM Serving

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OVERVIEW OF THE LLM STRUCTURE

Basic workflow of LLM inference



Token	Index
the	1
mouse	2
ran	3
up	4
clock	5
down	6

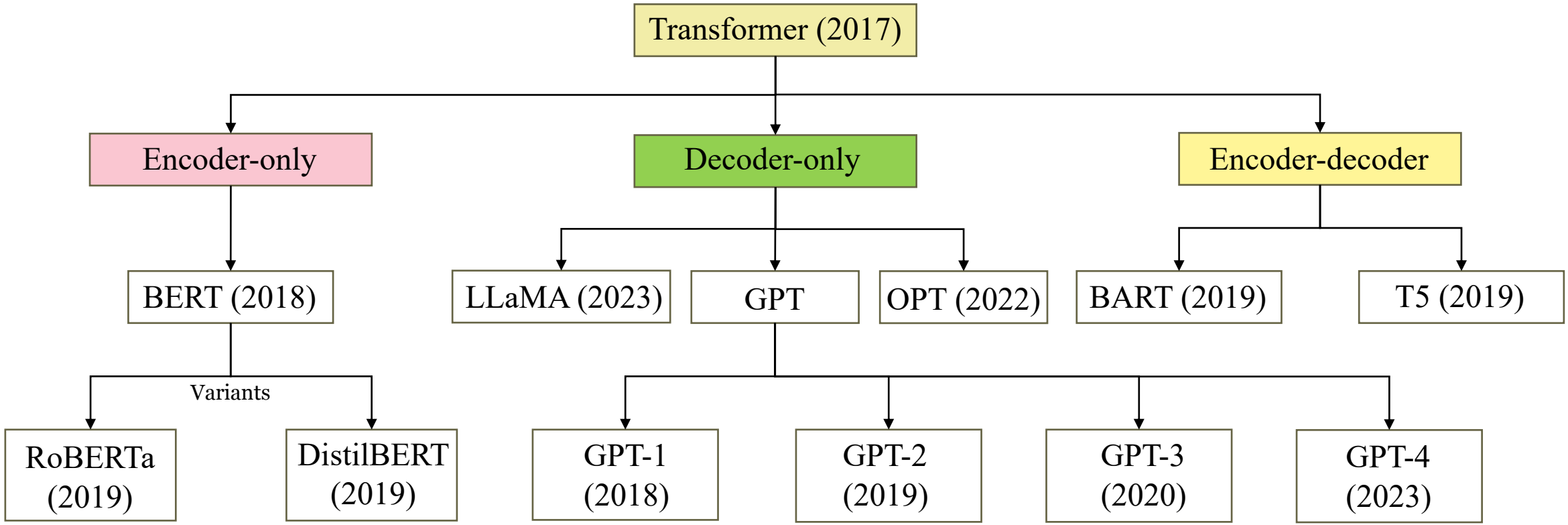
The mouse ran down → [1, 2, 3, 6] → Embedding Layer (output dim = 4) →

[[0.236, -0.141, 0.000, 0.045], ← the
[0.006, 0.652, 0.270, -0.556], ← mouse
[0.305, 0.569, -0.028, 0.496], ← ran
[0.466, -0.326, 0.884, 0.007]] ← down

Input Sequence | Vocabulary | Token index | Input embeddings to the transformer

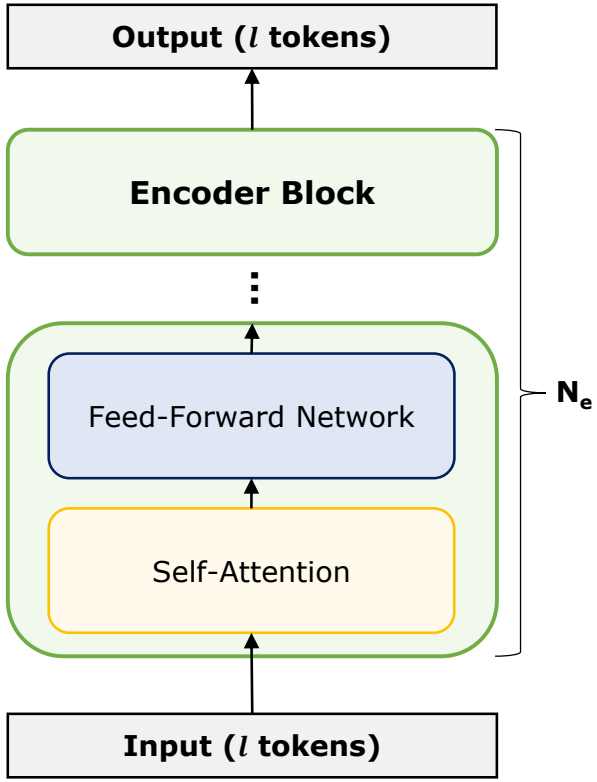
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Three major variants of Transformer networks

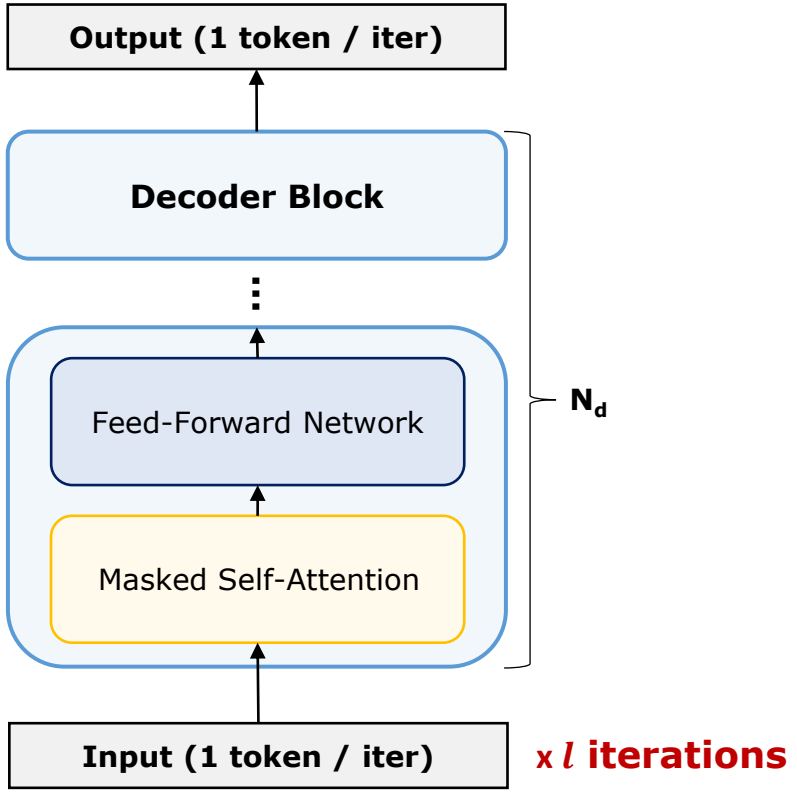


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Parallel processing vs. Autoregressive generation



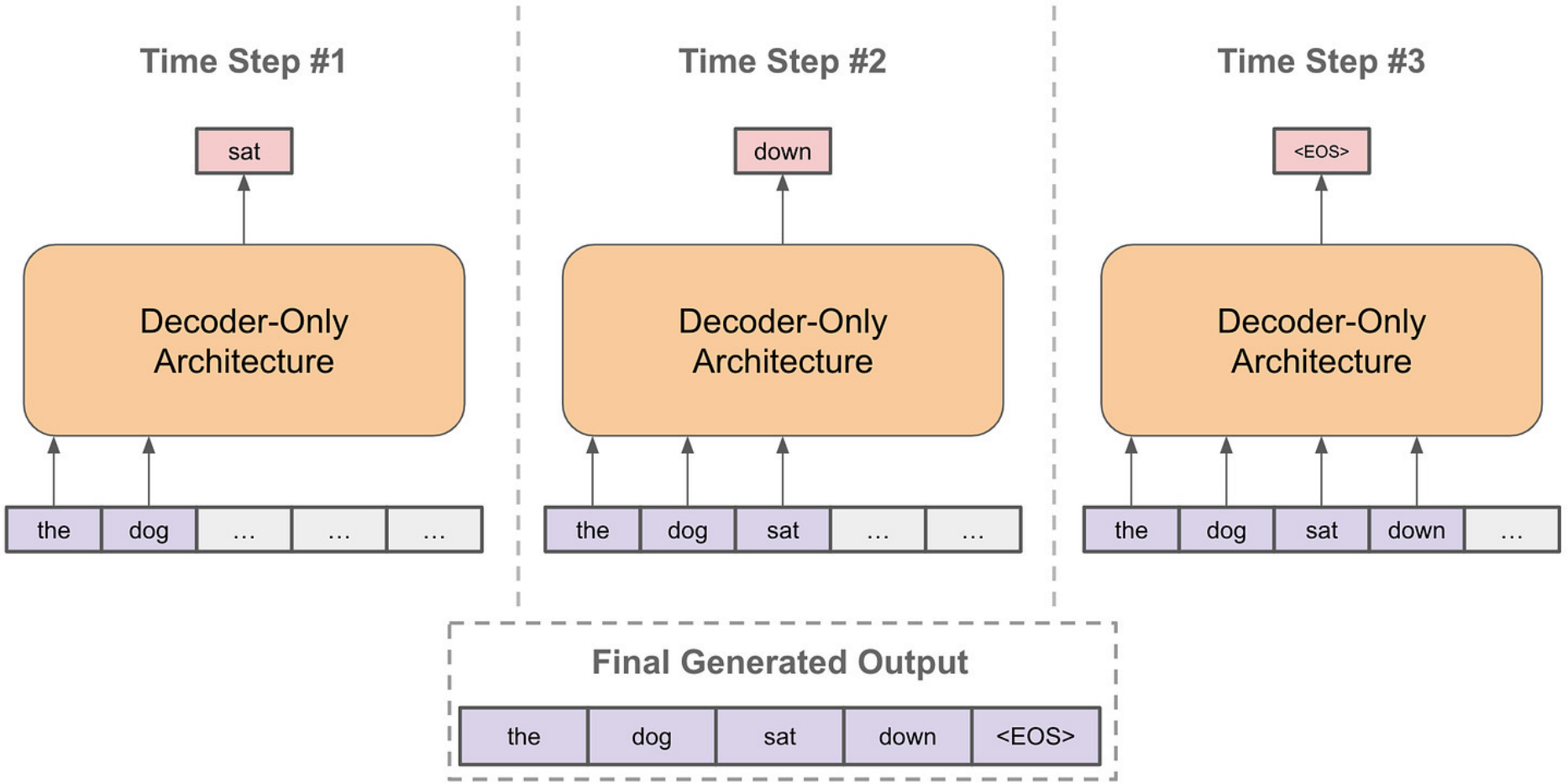
(a) Encoder-Only



(b) Decoder-Only

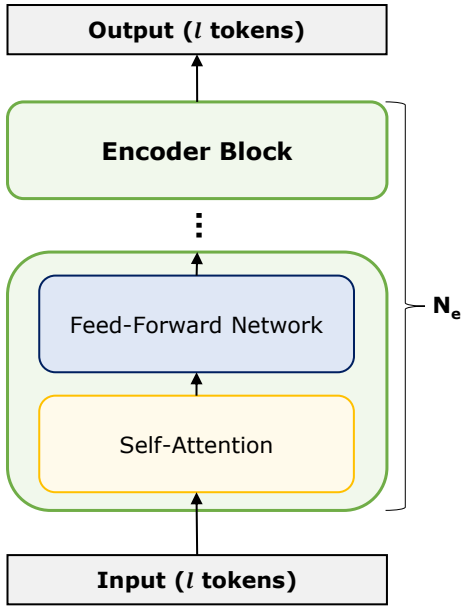
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Autoregressive Inference



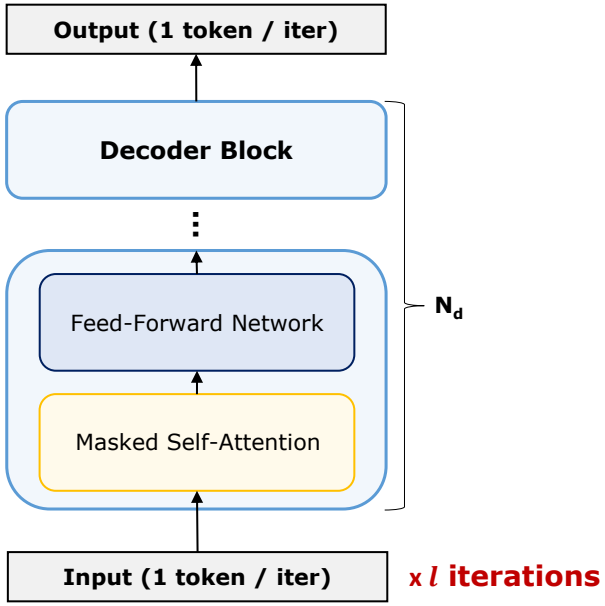
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Bidirectional Attention vs. Unidirectional Attention



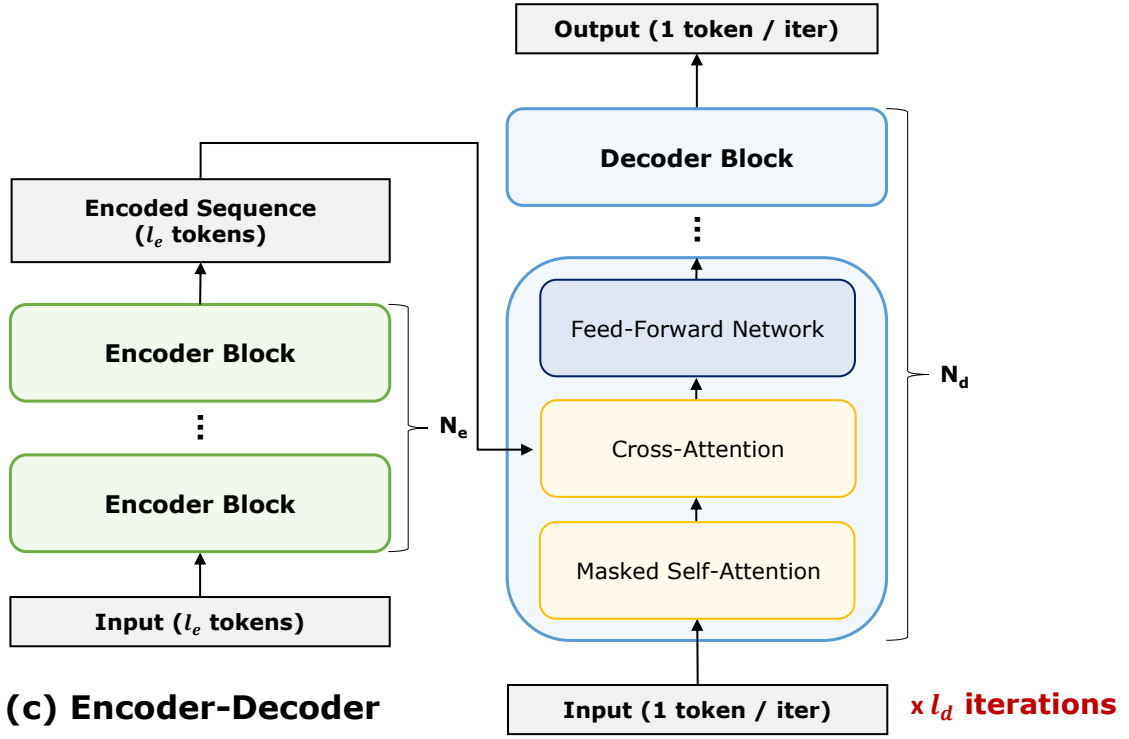
(a) Encoder-Only

Bidirectional Understanding



(b) Decoder-Only

Unidirectional Understanding

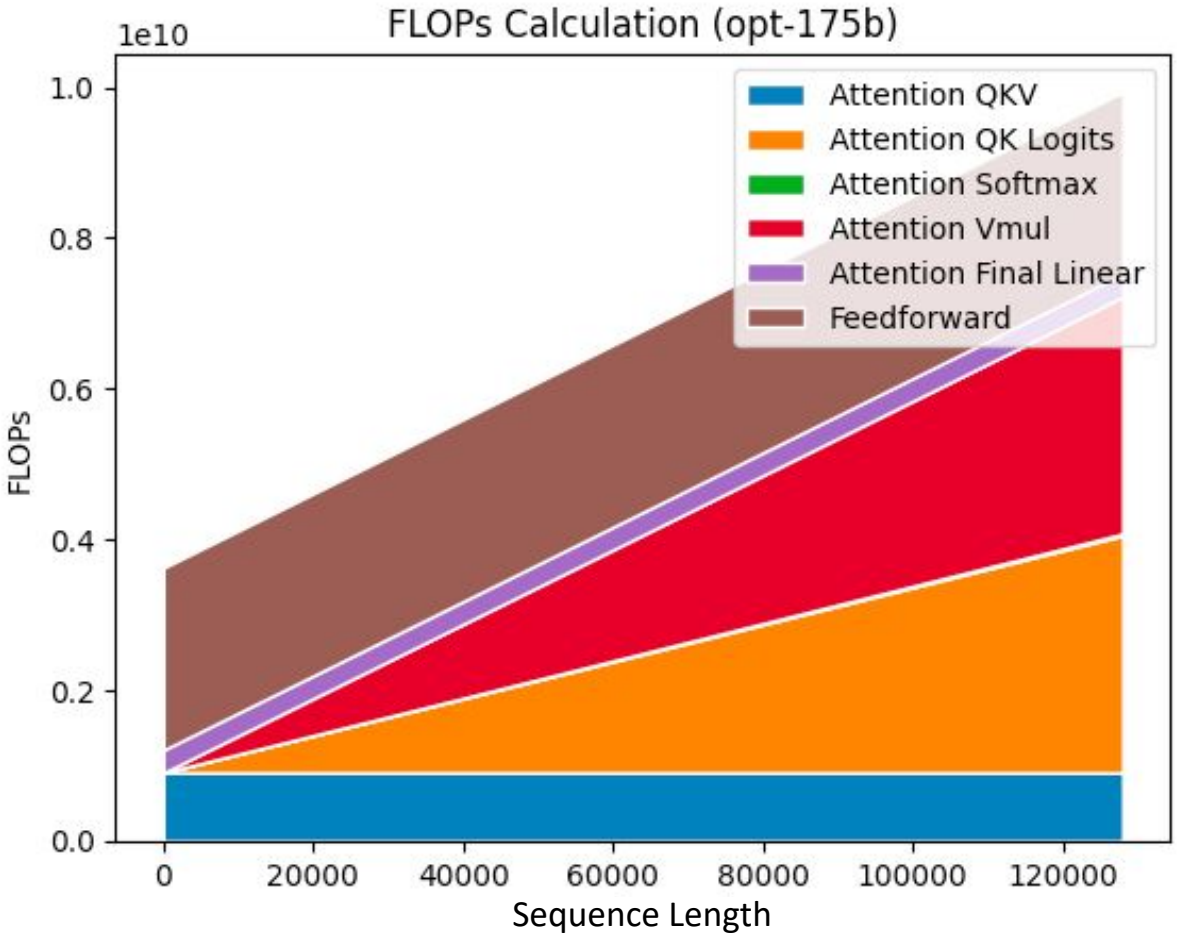


(c) Encoder-Decoder

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BASIC COMPONENTS: SELF-ATTENTION AND FFN

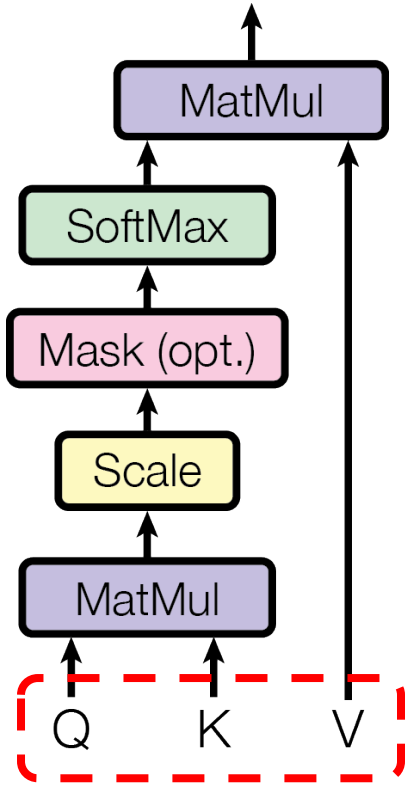
Serving Overhead Breakdown



Matrix Multiplications in self-attention and FFN take up a major portion of calculations in LLM inference.

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Self-Attention



Step 1: Creating Query, Key and Value vectors

Input Embedding $x = [a \quad b \quad c \quad d]^T$

a, b, c and d are column vectors with the length of embedding size.

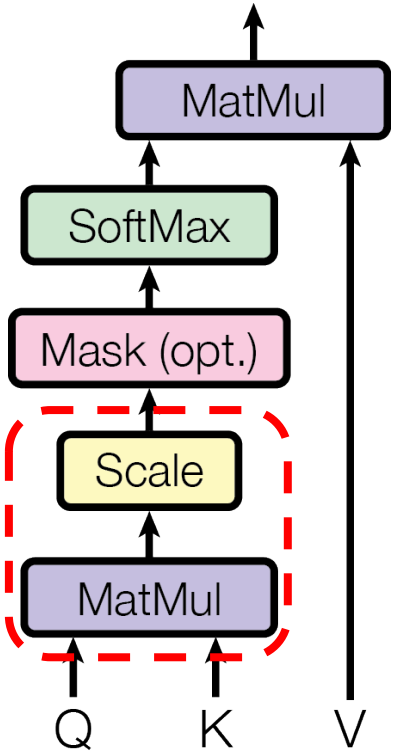
Query $Q = x \cdot W_Q = [a^Q \quad b^Q \quad c^Q \quad d^Q]^T$

Key $K = x \cdot W_K = [a^K \quad b^K \quad c^K \quad d^K]^T$

Value $V = x \cdot W_V = [a^V \quad b^V \quad c^V \quad d^V]^T$

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Self-Attention



Step 2: Calculate score of tokens against each other

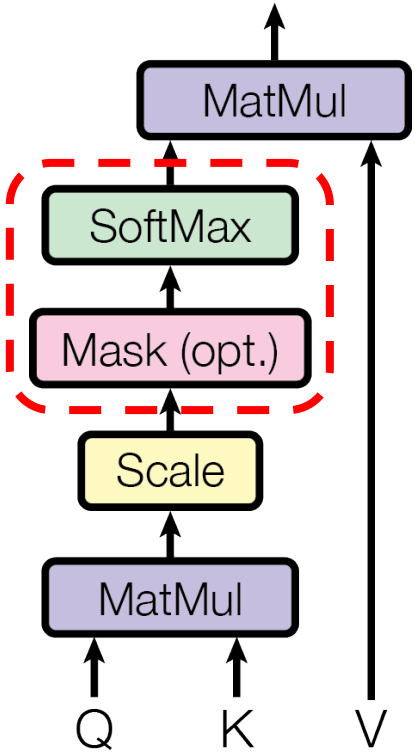
Query $Q = x \cdot W_Q = [a^Q \quad b^Q \quad c^Q \quad d^Q]^T$

Key $K = x \cdot W_K = [a^K \quad b^K \quad c^K \quad d^K]^T$

$$QK^T = \begin{bmatrix} (a^Q)^T a^K & (a^Q)^T b^K & (a^Q)^T c^K & (a^Q)^T d^K \\ (b^Q)^T a^K & (b^Q)^T b^K & (b^Q)^T c^K & (b^Q)^T d^K \\ (c^Q)^T a^K & (c^Q)^T b^K & (c^Q)^T c^K & (c^Q)^T d^K \\ (d^Q)^T a^K & (d^Q)^T b^K & (d^Q)^T c^K & (d^Q)^T d^K \end{bmatrix}$$

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Self-Attention



Step 3: Masking scores from the succeeding tokens

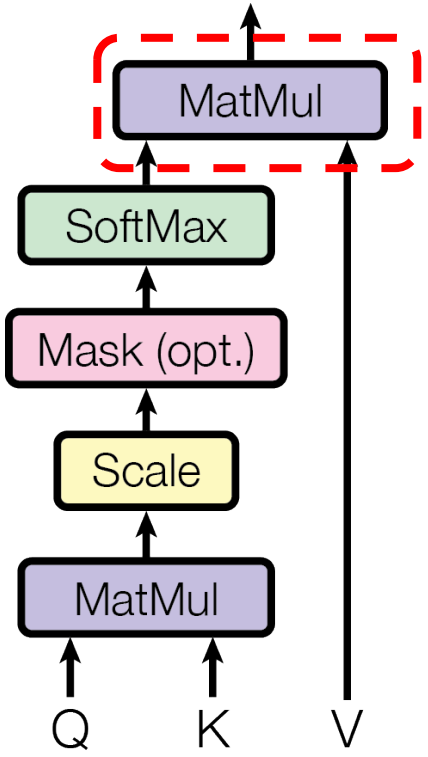
$$QK^T = \begin{bmatrix} (a^Q)^T a^K & (a^Q)^T b^K & (a^Q)^T c^K & (a^Q)^T d^K \\ (b^Q)^T a^K & (b^Q)^T b^K & (b^Q)^T c^K & (b^Q)^T d^K \\ (c^Q)^T a^K & (c^Q)^T b^K & (c^Q)^T c^K & (c^Q)^T d^K \\ (d^Q)^T a^K & (d^Q)^T b^K & (d^Q)^T c^K & (d^Q)^T d^K \end{bmatrix} \quad M = \begin{bmatrix} 0 & -\infty & -\infty & -\infty \\ 0 & 0 & -\infty & -\infty \\ 0 & 0 & 0 & -\infty \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$QK^T + M = \begin{bmatrix} (a^Q)^T a^K & -\infty & -\infty & -\infty \\ (b^Q)^T a^K & (b^Q)^T b^K & -\infty & -\infty \\ (c^Q)^T a^K & (c^Q)^T b^K & (c^Q)^T c^K & -\infty \\ (d^Q)^T a^K & (d^Q)^T b^K & (d^Q)^T c^K & (d^Q)^T d^K \end{bmatrix} \quad \text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

$$\text{softmax}(QK^T + M) = \begin{bmatrix} (a^Q)^T a^K & 0 & 0 & 0 \\ s((b^Q)^T a^K & (b^Q)^T b^K) & 0 & 0 \\ s((c^Q)^T a^K & (c^Q)^T b^K & (c^Q)^T c^K) & 0 \\ s((d^Q)^T a^K & (d^Q)^T b^K & (d^Q)^T c^K & (d^Q)^T d^K) \end{bmatrix}$$

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Self-Attention



Step 4: Dot production with the Value vector

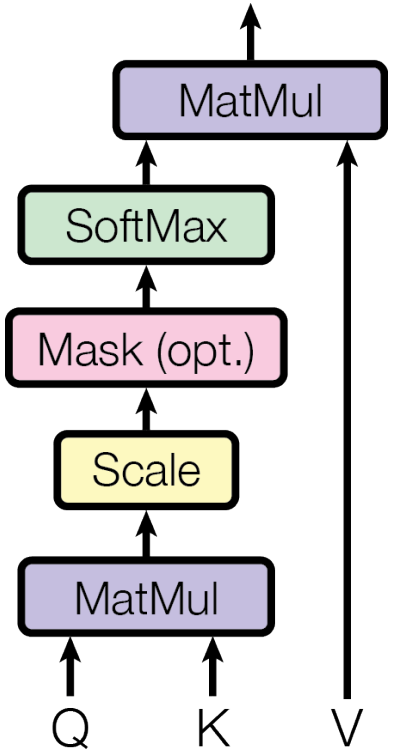
$$\text{softmax}(QK^T + M) = \begin{bmatrix} (a^Q)^T a^K & 0 & 0 & 0 \\ s((b^Q)^T a^K & (b^Q)^T b^K) & 0 & 0 \\ s((c^Q)^T a^K & (c^Q)^T b^K & (c^Q)^T c^K) & 0 \\ s((d^Q)^T a^K & (d^Q)^T b^K & (d^Q)^T c^K & (d^Q)^T d^K) \end{bmatrix}$$

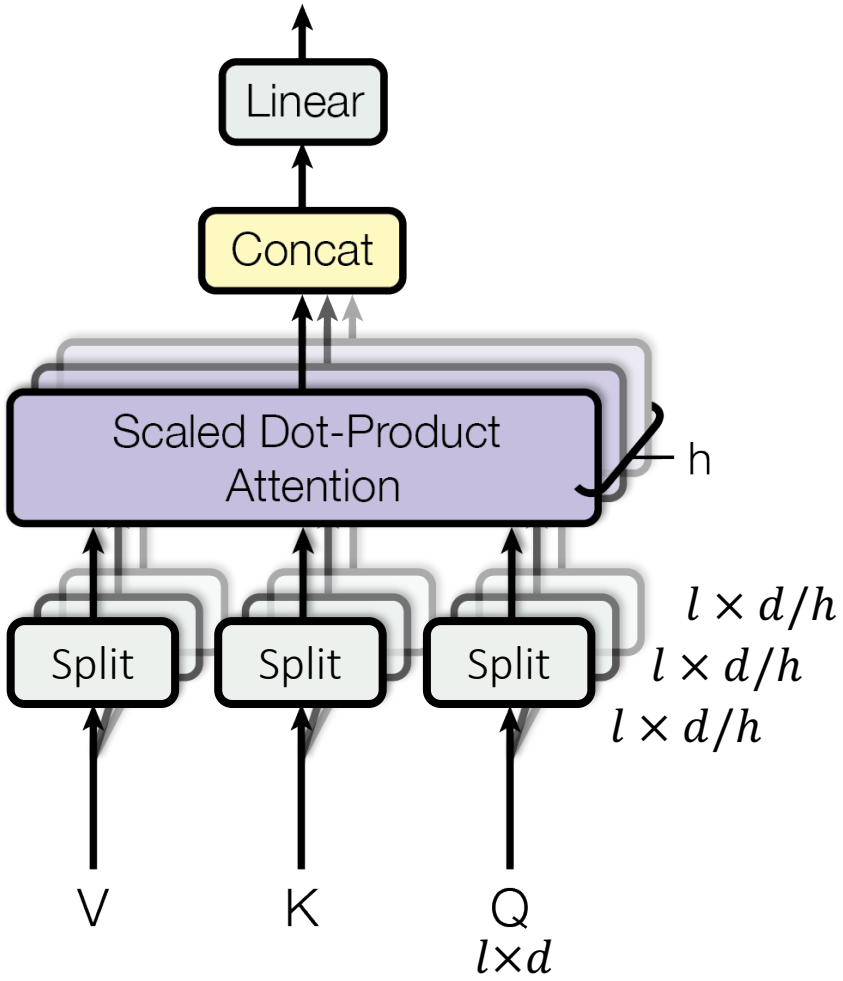
$$\text{softmax}(QK^T + M) \cdot V = \begin{bmatrix} (a^Q)^T a^K & 0 & 0 & 0 \\ s((b^Q)^T a^K & (b^Q)^T b^K) & 0 & 0 \\ s((c^Q)^T a^K & (c^Q)^T b^K & (c^Q)^T c^K) & 0 \\ s((d^Q)^T a^K & (d^Q)^T b^K & (d^Q)^T c^K & (d^Q)^T d^K) \end{bmatrix} \begin{bmatrix} (a^V)^T \\ (b^V)^T \\ (c^V)^T \\ (d^V)^T \end{bmatrix}$$

$$\xRightarrow{\text{If embedding size=2}} \begin{bmatrix} (a^Q)^T a^K a_1^V & (a^Q)^T a^K a_2^V \\ \left((b^Q)^T a^K\right)' a_1^V + \left((b^Q)^T b^K\right)' b_1^V & \left((b^Q)^T a^K\right)' a_2^V + \left((b^Q)^T b^K\right)' b_2^V \\ \left((c^Q)^T a^K\right)' a_1^V + \left((c^Q)^T b^K\right)' b_1^V + \left((c^Q)^T c^K\right)' c_1^V & \left((c^Q)^T a^K\right)' c_2^V + \left((c^Q)^T b^K\right)' b_2^V + \left((c^Q)^T c^K\right)' c_2^V \\ \left((d^Q)^T a^K\right)' a_1^V + \left((d^Q)^T b^K\right)' b_1^V + \left((d^Q)^T c^K\right)' c_1^V + \left((d^Q)^T d^K\right)' d_1^V & \left((d^Q)^T a^K\right)' a_2^V + \left((d^Q)^T b^K\right)' b_2^V + \left((d^Q)^T c^K\right)' c_2^V + \left((d^Q)^T d^K\right)' d_2^V \end{bmatrix}$$

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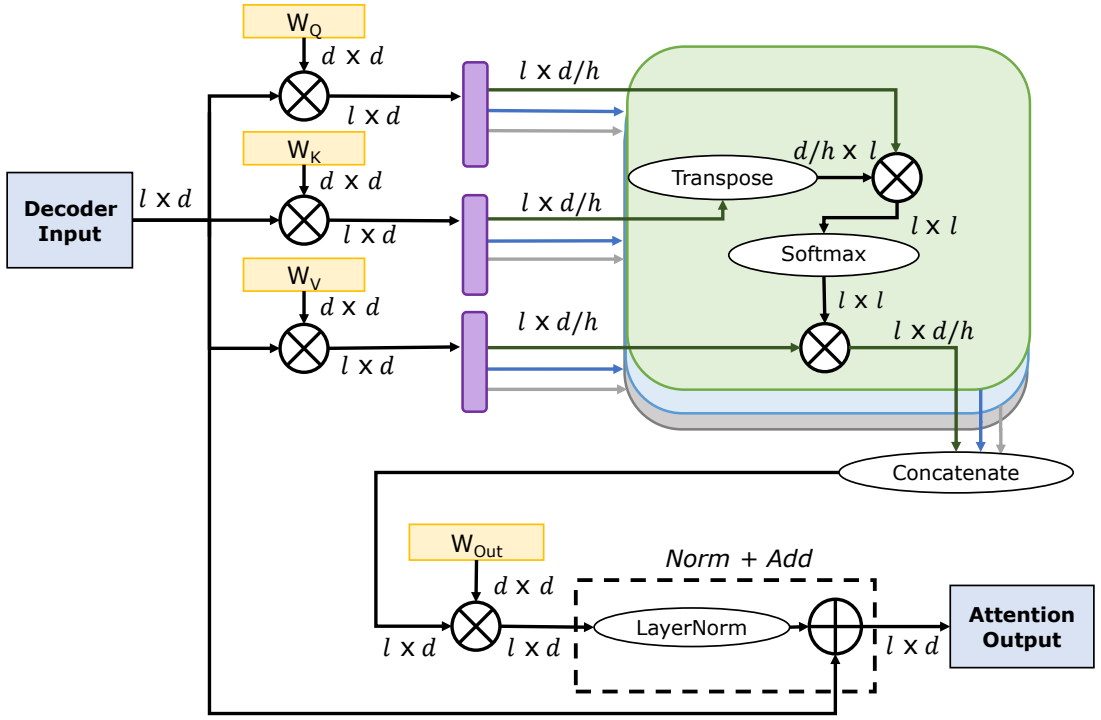
Multi-Head Self-Attention





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Self-Attention

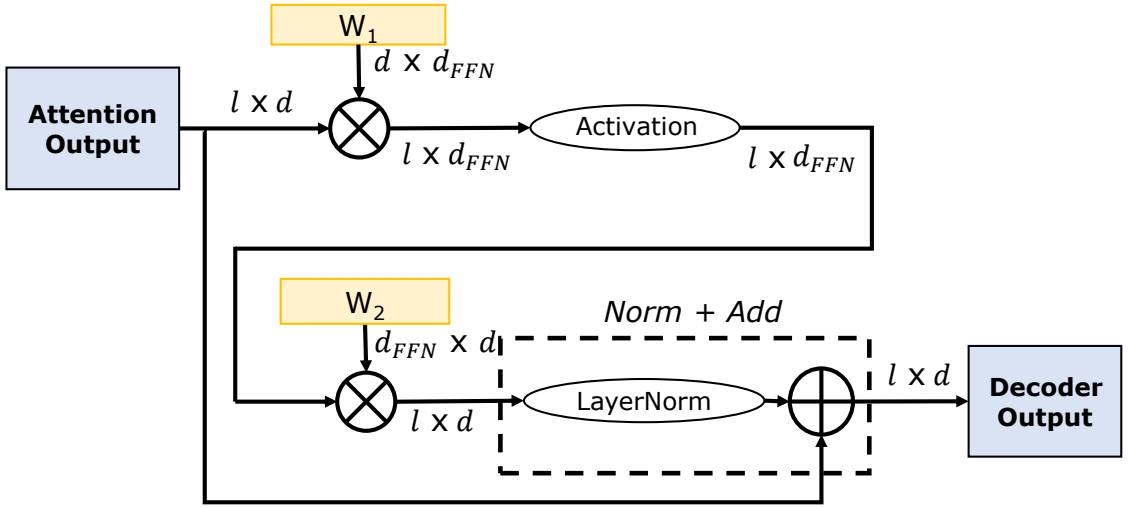


$$Q = x \cdot W_Q$$
$$K = x \cdot W_K$$
$$V = x \cdot W_V$$

$$\text{Attention}(Q, K, V) = x + \text{LayerNorm}\left(\text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V\right)$$

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Feed-Forward Network (FFN)



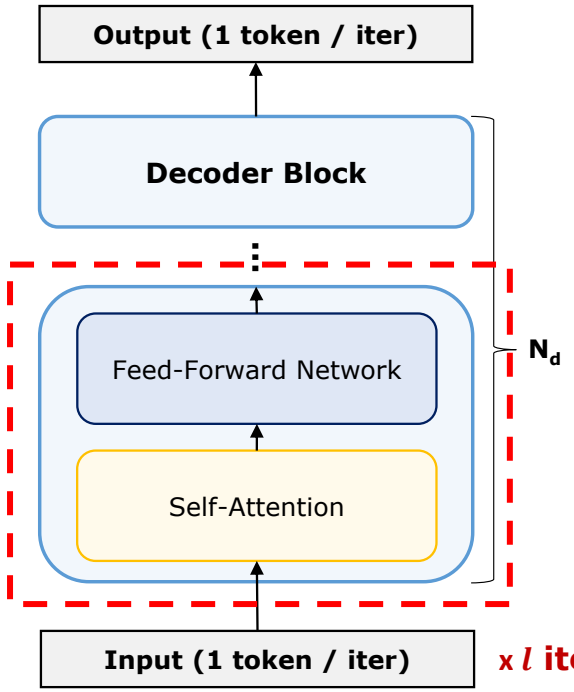
W_1 and W_2 projections: matrix multiplication between the input tensors and weight matrix.

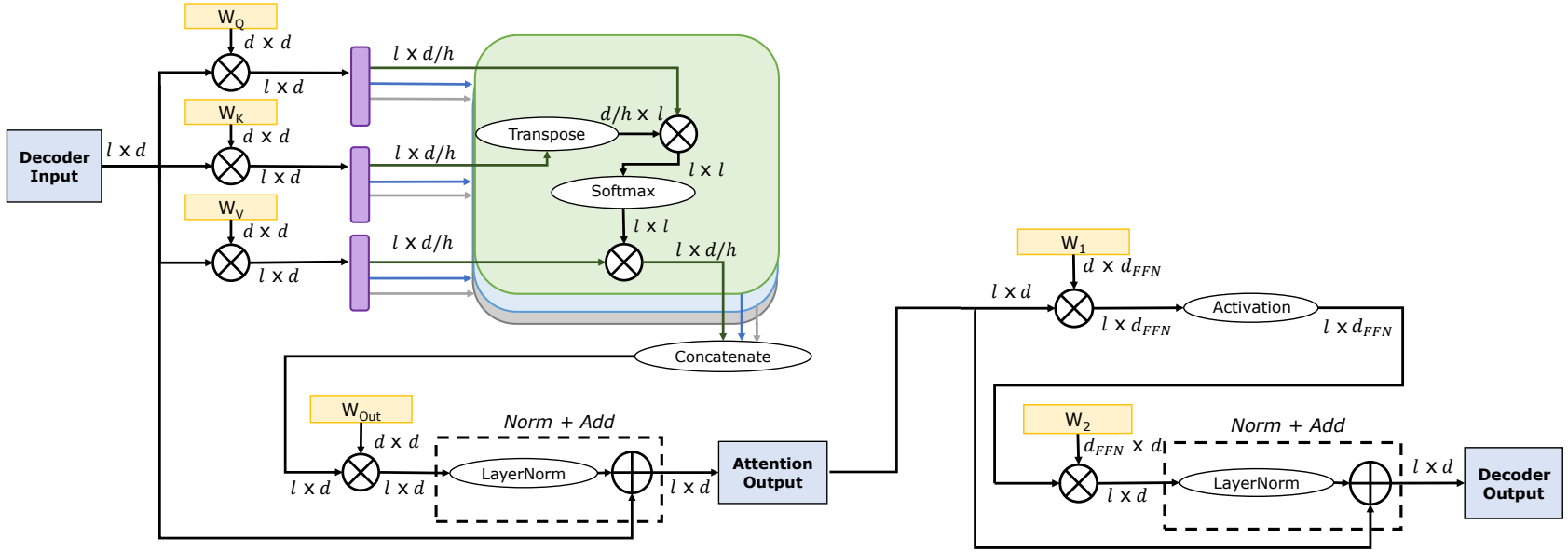
$$l\left\{\underbrace{\square}_{d}\right. \cdot \left.\underbrace{\square}_{d_{FFN}}\right\} d = \left.\underbrace{\square}_{d_{FFN}}\right\} l$$

$$l\left\{\underbrace{\square}_{d_{FFN}}\right. \cdot \left.\underbrace{\square}_{d}\right\} d_{FFN} = \left.\underbrace{\square}_{d}\right\} l$$

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Put them together



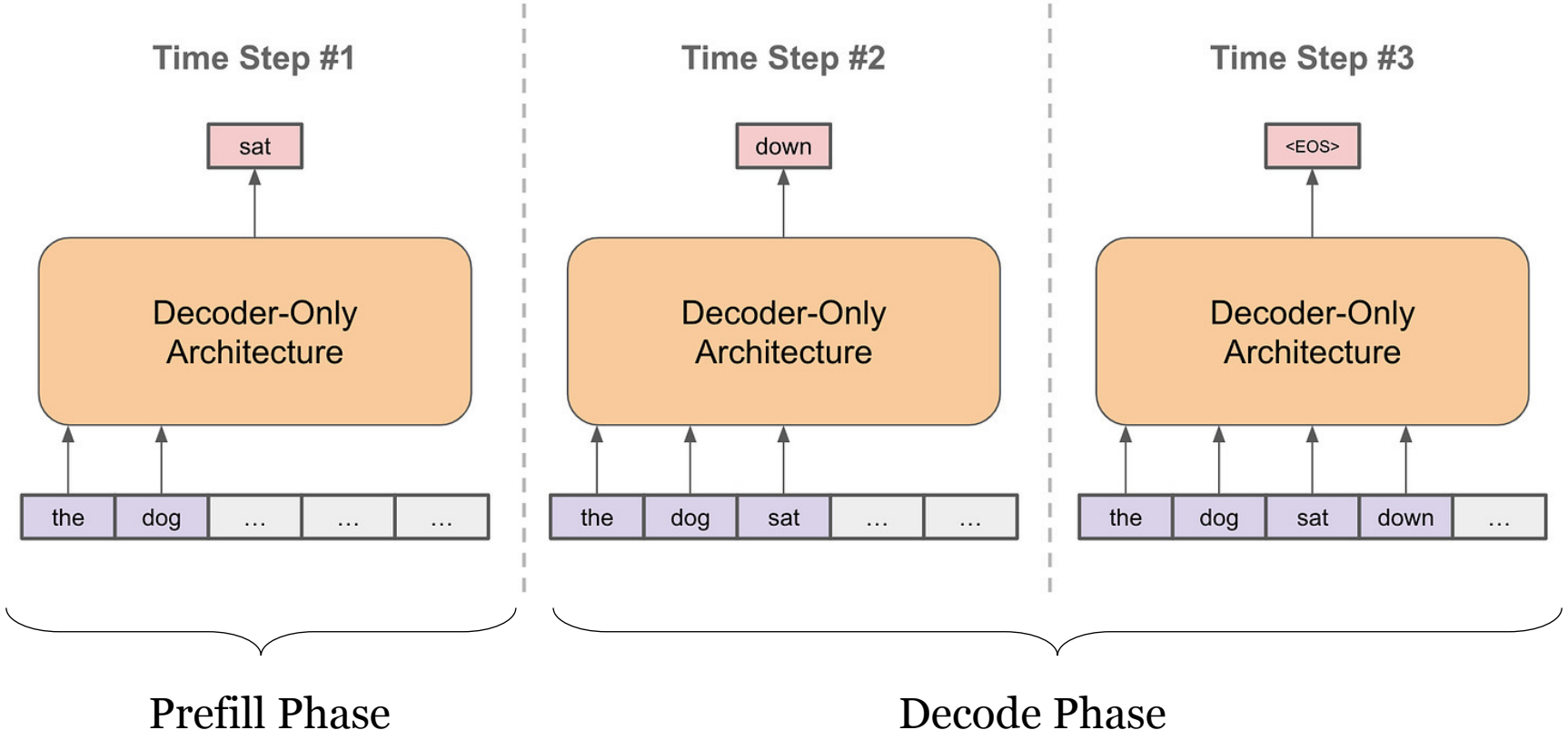


Model Name	n_{params}	n_{layers}	d	h
GPT-3 2.7B	2.7B	32	2560	32
GPT-3 6.7B	6.7B	32	4096	32
GPT-3 13B	13.0B	40	5140	40
GPT-3 175B or "GPT-3"	175.0B	96	12288	96

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SERVING PROCESS OF DECODER-ONLY LLM: PREFILL VS. DECODE

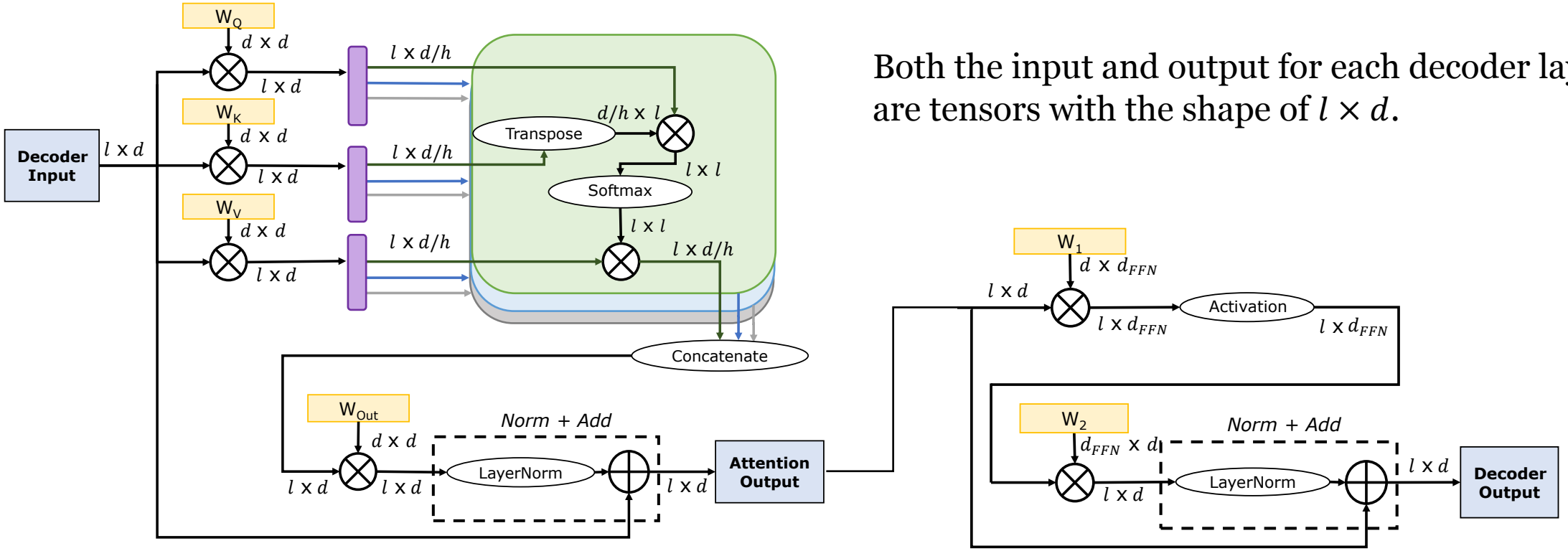
Prefill and Decode Phases



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Prefill Phase

The first iteration of the serving, receiving input prompt and generating the first output token.



Both the input and output for each decoder layer are tensors with the shape of $l \times d$.

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Decode Phase: Key-Value Caches

The subsequent token generation iterations after the first one. All input and output tokens are used for self-attention in the decode phase.

$$x = [a \quad b \quad c \quad d \quad e]^T$$

$$K = x \cdot W_K = [a^K \quad b^K \quad c^K \quad d^K \quad e^K]^T$$

$$V = x \cdot W_V = [a^V \quad b^V \quad c^V \quad d^V \quad e^V]^T$$



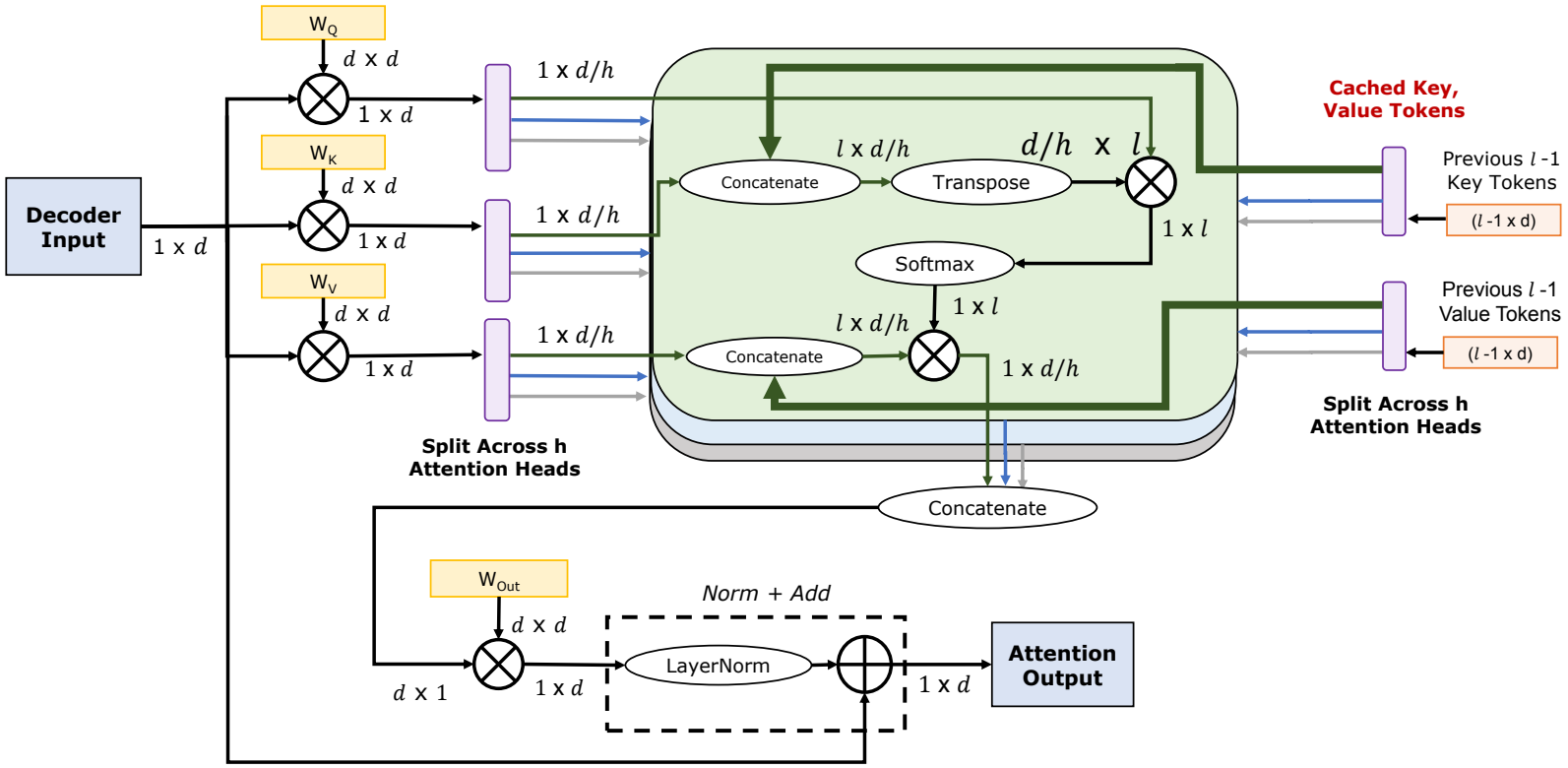
Calculated in the last iteration

New value in current iteration

Cache K-V matrices generated in each iteration and reuse them in the following iterations.

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Decode Phase: Workflow



1. Input tensor with the shape of $1 \times d$ only includes information for the last generated token

2. Previously cached key-value data is concatenated into the projection results.

3. Concatenated data follows the same calculation as the prefill phase, except that the output length is only one.

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Prefill vs. Decode

Computation and memory comparison between prefill and decode phases

Model Configuration (GPT3): $d = 12288, d_{\text{FFN}} = 49152, h = 96, n = 96$, Data bitwidth of weights: 16 bit
Workload: Number of tokens: 1000

Computation Workload for Each Iteration

Multi-Head Attention

of Floating-point operations

Prefill	Decode
2.5×10^{14}	1.2×10^{11}

Feed-Forward Network

of Floating-point operations

Prefill	Decode
4.6×10^{14}	2.3×10^{11}

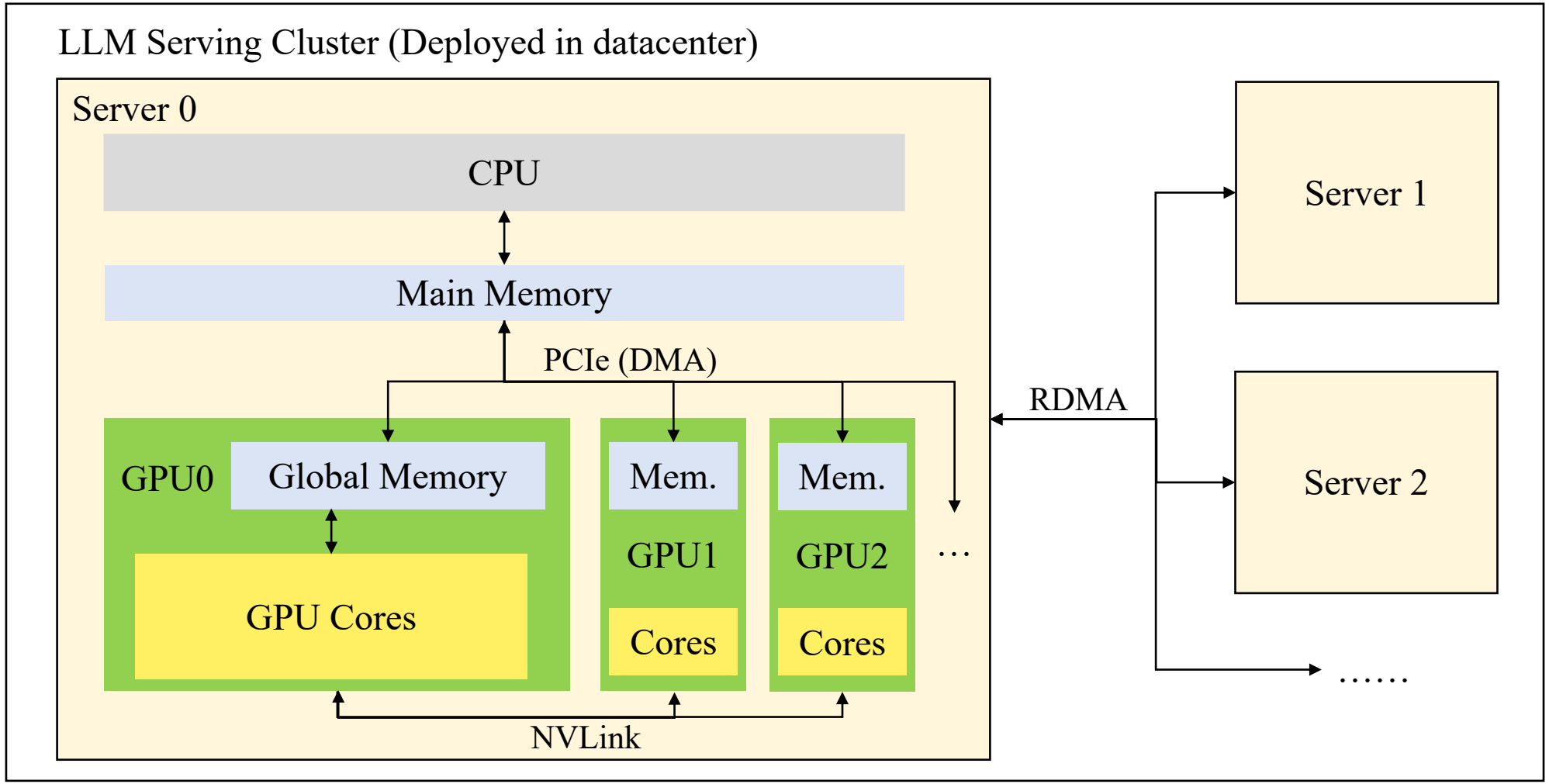
K-V Cache Size for Decoding: 4.39 GB

Prefill phase is compute-intensive while decode phase is memory-intensive.

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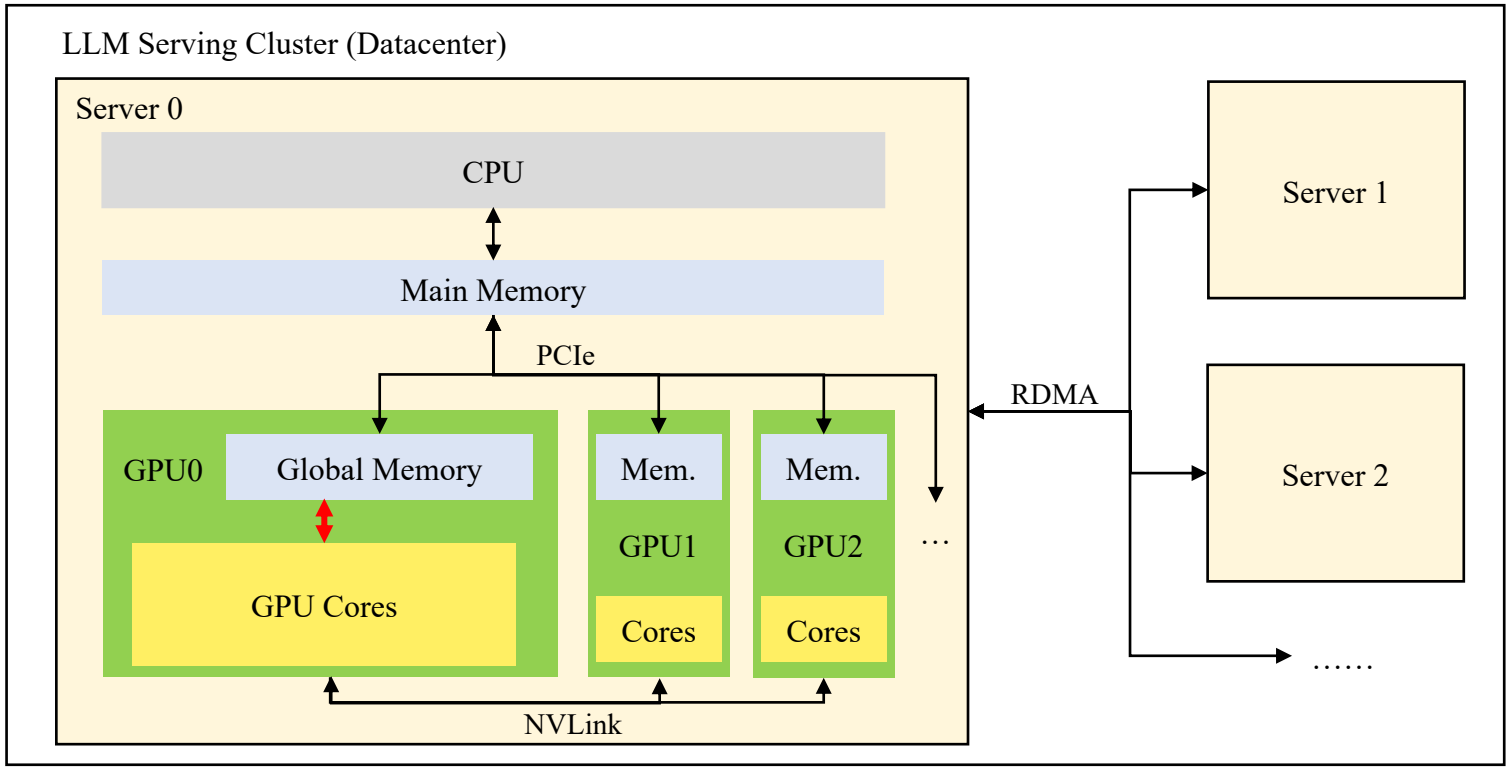
INFRASTRUCTURE FOR LLM SERVING

Distributed Serving System Overview



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Data Transmission Bandwidth and Memory Size



Intra-server Connections (NVIDIA DGX A100**):**

PCIe 4.0: 64GB/s

NVLink: 600GB/s

GPU Memory Bandwidth (HBM2): Around 1000-2000 GB/s

Inter-server Connections:
RDMA: 10-50 GB/s

Memory Size
Main Memory Size: Up to several TB
Global Memory Size: Less than 100 GB
Reference: Model Size of GPT 3 is 326GB

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COMMON CHALLENGES IN LLM SERVING

Common Challenges in LLM Serving

Complex and dynamic scheduling

1. Unpredictable output length
2. Autoregressive inference manner makes it difficult to batch jobs together

Memory constraint

1. Limited GPU memory against large model size
2. Large cache size in long context processing
3. Insufficient inter-device transmission bandwidth

Prefill vs. Decode

1. Batching between prefill and decode jobs is non-trivial
2. Prefill and decode jobs have different requirement over computation and memory

Compound AI and agentic systems

1. Diverse components increase the complexity of system
2. Varying and uncertain workflows require robust system design

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